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Capture the Flag 4: The Student List

Flag 0 — Create Your Feature Branch

```
WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/data-ui)
$ git checkout -b feature/student-list
Switched to a new branch 'feature/student-list'

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-list)
$ gi branch
bash: gi: command not found

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-list)
$ git branch
  feature/data-ui
  feature/exam
  feature/exam_2
* feature/student-list
  feature/widgets
  main
```

🚩 Flag 1 — Build Your Student Data

Objective: Store multiple students inside a Dart `List`.

Research what a Dart `List` is and why it is useful when handling multiple pieces of similar data.

Instead of creating one student at a time with separate variables, create a collection that stores at least five students.

Each student should contain information such as:

Image

Name

Course

Year Level

Age

Hobby

Your data should exist separately from your UI widgets.

Screenshot:

```
class Profile {
    final String image;
    final String? name;
    final String? course;
    final String? yearLevel;
    final int? age;
    final String? hobby;

    const Profile({
        required this.image,
        this.name,
        this.course,
        this.yearLevel,
        this.age,
        this.hobby,
    });
}

const List<Profile> profiles = [
    Profile(
        image: "assets/satorugojo.jpg",
        name: "Satoru Gojo",
        course: "BSIT",
        yearLevel: "3rd Year",
        age: 20,
        hobby: "Eating Sweets",
    ),
    Profile(
        image: "assets/sugurugeto.jpg",
        name: "Suguru Geto",
        course: "BSIT",
        yearLevel: "3rd Year",
        age: 19,
        hobby: "Reading",
    ),
    Profile(
        image: "assets/chosokamo.jpg",
        name: "Choso Kamo",
        course: "BSIT",
        yearLevel: "4th Year",
        age: 25,
        hobby: "Spending time with brothers",
    ),
    Profile(
        image: "assets/sukuna.jpg",
        name: "Ryomen Sukuna",
        course: "BSIT",
        yearLevel: "4th Year",
        age: 30,
        hobby: "Eating",
    ),
    Profile(
        image: "assets/makizenin.jpg",
        name: "Maki Zenin",
        course: "BSIT",
        yearLevel: "1st Year",
        age: 20,
        hobby: "Crushing Cans",
    ),
];
```

🚩 Flag 2 — The Copy-Paste Trap

You already know how to create a Student Card.

You created five student cards manually.

You duplicated the same card design until all five students appear.

Stop.

Imagine having:

50 students.

500 students.

5,000 students.

Would copying the same widget over and over still be practical?

This flag is intentionally making you feel the problem before solving it.

Screenshot:

```
//, // AppBar
body: SingleChildScrollView(
  child: Column(
    crossAxisAlignment: CrossAxisAlignment.center,
    children: [
      const SizedBox(height: 20),

      Padding(
        padding: const EdgeInsets.only(
          left: 20,
          right: 20,
          bottom: 20,
        ), // EdgeInsets.only
        child: Card(
          elevation: 12,
          color: const Color.fromARGB(255, 253, 253, 253),
          shape: RoundedRectangleBorder(
            borderRadius: BorderRadius.circular(25),
          ), // RoundedRectangleBorder
          child: Padding(
            padding: const EdgeInsets.all(30.0),
            child: Column(
              mainAxisAlignment: MainAxisAlignment.min,
              mainAxisAlignment: MainAxisAlignment.center,
              crossAxisAlignment: CrossAxisAlignment.center,
              children: [
                Image.asset(
                  "assets/satorugojo.jpg",
                  width: 140,
```

```
height: 140,
                ), // Image.asset
                const SizedBox(height: 20),
              ], // Row
            ), // Row
          ), // Row
        ), // Row
      ), // Row
    ], // Column
  ), // SingleChildScrollView
), // Scaffold
), // MaterialApp
), // runApp
```


Research Flutter's `ListView`.

Replace your previous layout with a scrolling layout so users can browse through all of your student cards naturally.

Your application should now allow continuous vertical scrolling.

Test

The user should be able to:

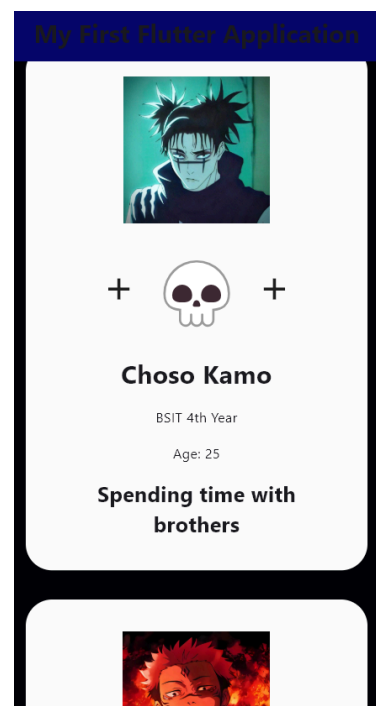
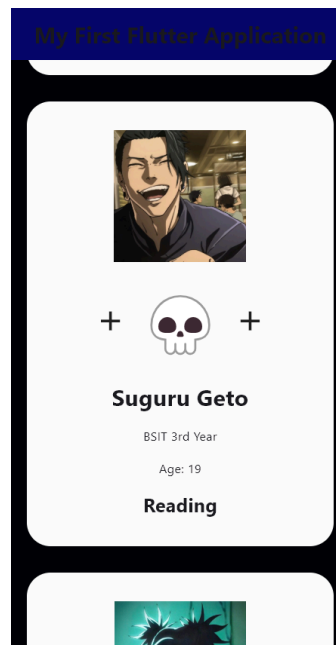
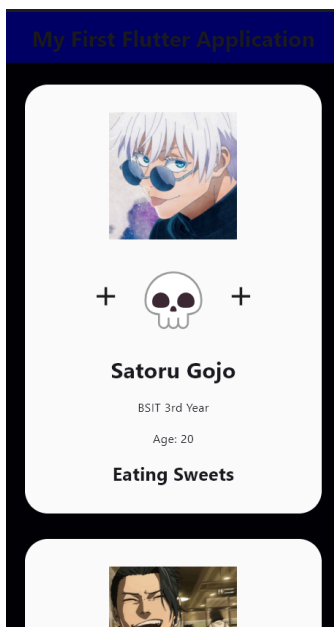
Scroll from the first student

Continue scrolling

Reach the last student without resizing the screen

Screenshot:

```
body: ListView(  
  children: [  
    
```



🚩 Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

Research `ListView.builder`.

Instead of writing five separate Student Cards, generate them automatically from your student list.

Your UI should now create one card for each student in your list.

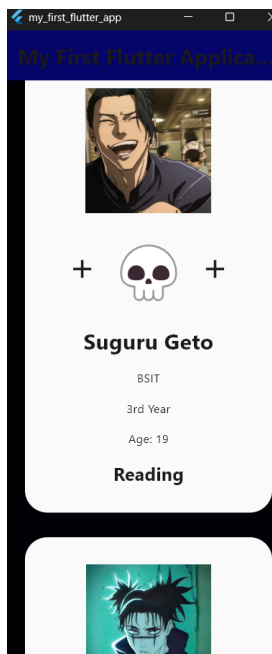
Test

Add one new student to your list.

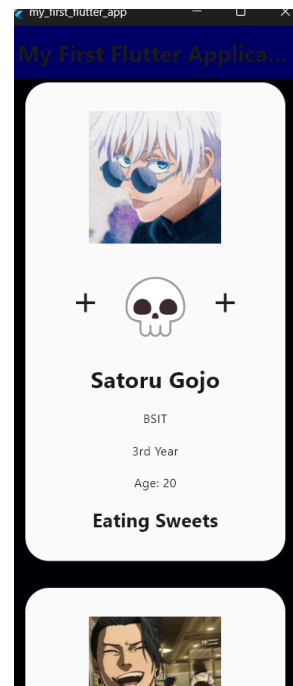
Without creating another card manually, your application should display the new student automatically.

Screenshot:

```
body: ListView.builder(  
  itemCount: profiles.length,  
  itemBuilder: (context, index) {  
    final profile = profiles[index];
```



```
const List<Profile> profiles = [  
  Profile(  
    image: "assets/satorugojo.jpg",  
    name: "Satoru Gojo",  
    course: "BSIT",  
    yearLevel: "3rd Year",  
    age: 20,  
    hobby: "Eating Sweets",  
  ),  
  Profile(  
    image: "assets/sugurugeto.jpg",  
    name: "Suguru Geto",  
    course: "BSIT",  
    yearLevel: "3rd Year",  
    age: 19,  
    hobby: "Reading",  
  ),  
  Profile(  
    image: "assets/chosokamo.jpg",  
    name: "Choso Kamo",  
    course: "BSIT",  
    yearLevel: "4th Year",  
    age: 25,  
    hobby: "Spending time with brothers",  
  ),  
  Profile(  
    image: "assets/sukuna.jpg",  
    name: "Ryomen Sukuna",  
    course: "BSIT",  
    yearLevel: "4th Year",  
    age: 30,  
    hobby: "Eating",  
  ),  
  Profile(  
    image: "assets/makizenin.jpg",  
    name: "Maki Zenin",  
    course: "BSIT",  
    yearLevel: "1st Year",  
    age: 20,  
    hobby: "Crushing Cans",  
  ),  
];
```



🚩 Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

Real student directories contain more than just names.

Add at least three new fields to every student.

Examples:

Student ID

Email

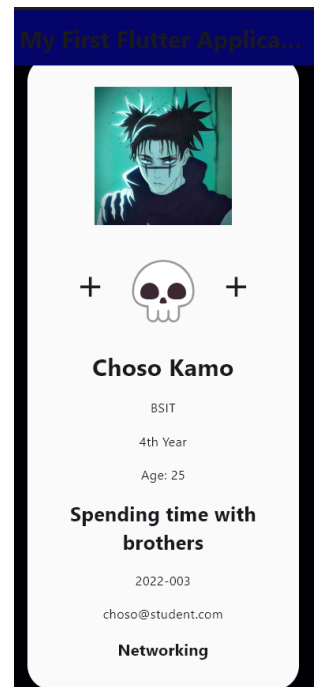
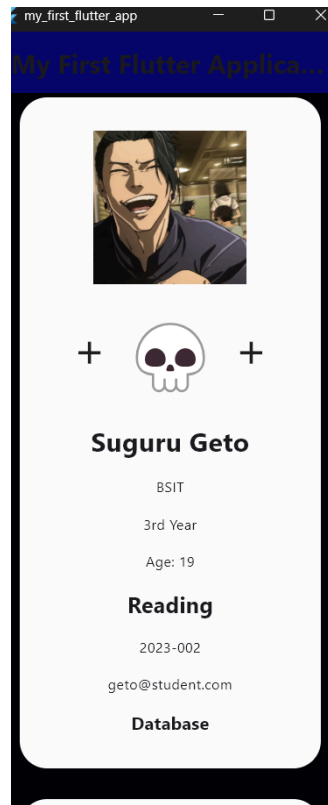
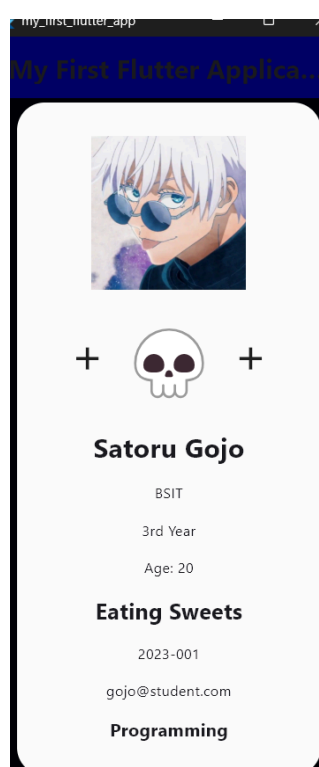
Favorite Subject

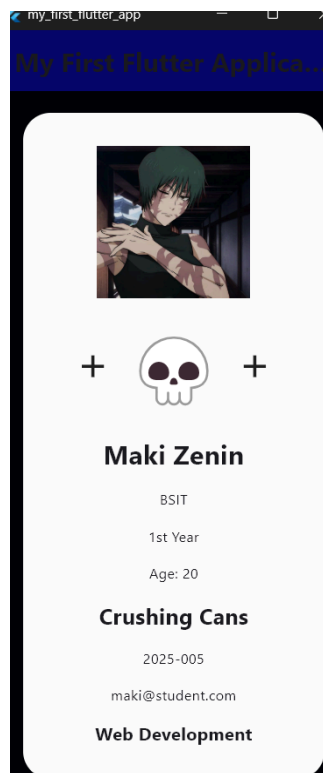
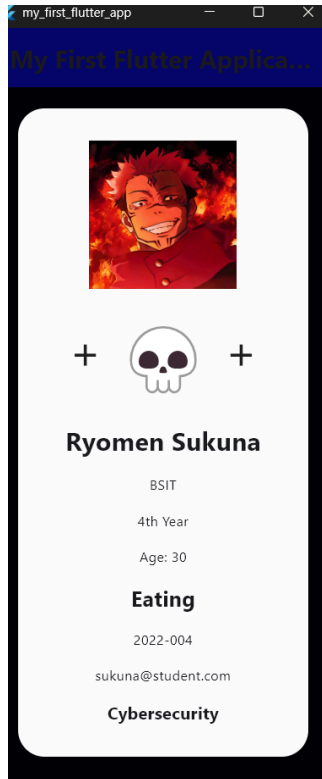
Update your Student Card so the new information appears consistently across every generated card.

Screenshot:

```
final String? email;  
final String? studentId;  
final String? email;  
final String? favoriteSubject;
```

```
this.studentId,  
this.email,  
this.favoriteSubject,  
).
```





```
const List<Profile> profiles = [
  Profile(
    image: "assets/satorugojo.jpg",
    name: "Satoru Gojo",
    course: "BSIT",
    yearLevel: "3rd Year",
    age: 20,
    hobby: "Eating Sweets",
    studentId: "2023-001",
    email: "gojo@student.com",
    favoriteSubject: "Programming",
  ),
  Profile(
    image: "assets/sugurugeto.jpg",
    name: "Suguru Geto",
    course: "BSIT",
    yearLevel: "3rd Year",
    age: 19,
    hobby: "Reading",
    studentId: "2023-002",
    email: "geto@student.com",
    favoriteSubject: "Database",
  ),
  Profile(
    image: "assets/chosokamo.jpg",
    name: "Choso Kamo",
    course: "BSIT",
    yearLevel: "4th Year",
    age: 25,
    hobby: "Spending time with brothers",
    studentId: "2022-003",
    email: "choso@student.com",
    favoriteSubject: "Networking",
  ),
];
```

```
Profile(
  image: "assets/sukuna.jpg",
  name: "Ryomen Sukuna",
  course: "BSIT",
  yearLevel: "4th Year",
  age: 30,
  hobby: "Eating",
  studentId: "2022-004",
  email: "sukuna@student.com",
  favoriteSubject: "Cybersecurity",
),
Profile(
  image: "assets/maki.jpg",
  name: "Maki Zenin",
  course: "BSIT",
  yearLevel: "1st Year",
  age: 20,
  hobby: "Crushing Cans",
  studentId: "2025-005",
  email: "maki@student.com",
  favoriteSubject: "Web Development",
),
```

🚩 Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty `List` gracefully.

Imagine your app just connected to a database.

The database returns:

```
[]
```

No students.

What should the user see?

Instead of displaying a broken or confusing screen, research how Flutter can display a different widget when a condition is met.

Create an empty state that clearly communicates that there are currently no students available.

Examples:

"No students found."

"Student list is empty."

Test

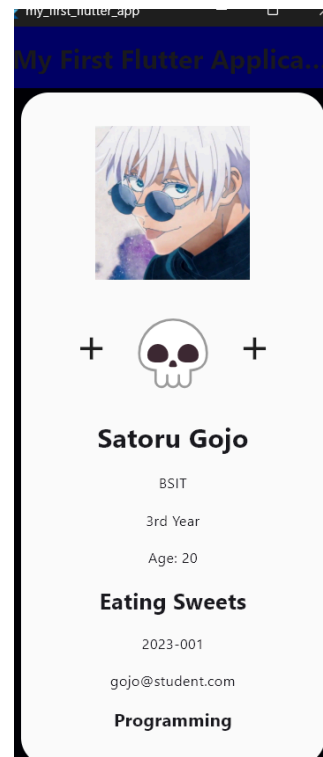
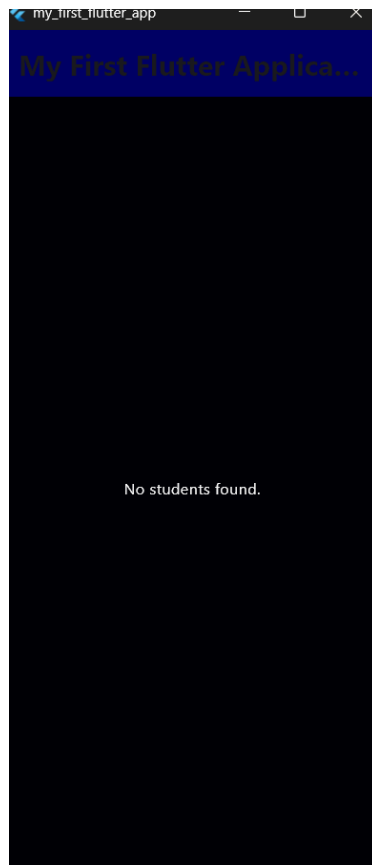
Display your student list normally.

Then temporarily make your list empty.

Your application should display the empty state instead of student cards.

Screenshot:

```
List<Profile> profiles = [];
```



Flag 7 — Sort the Roll Call

Objective: Change the order of your student list.

Class lists are rarely displayed randomly.

Research how Dart can rearrange items inside a [List](#).

Sort your students alphabetically by name.

Test

Before sorting:

Maria

John

Aaron

Sophia

After sorting:

Aaron

John

Maria

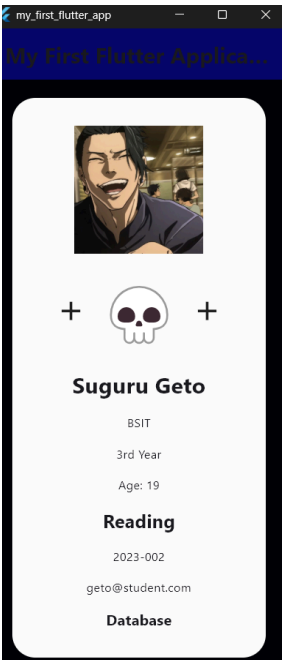
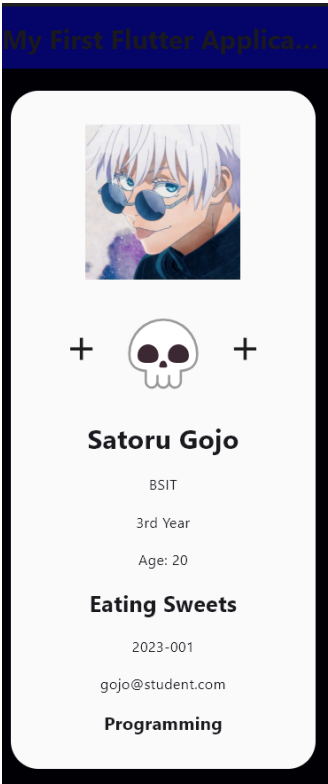
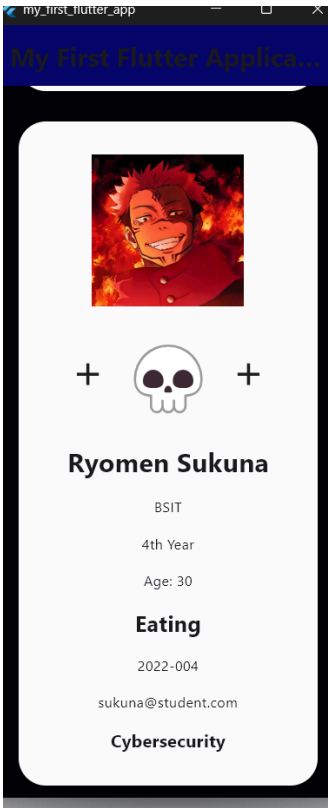
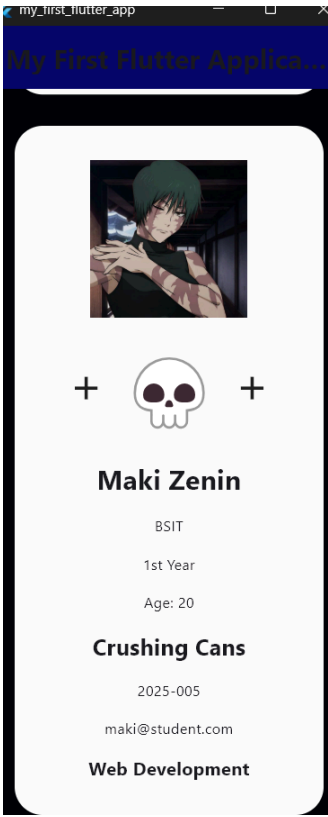
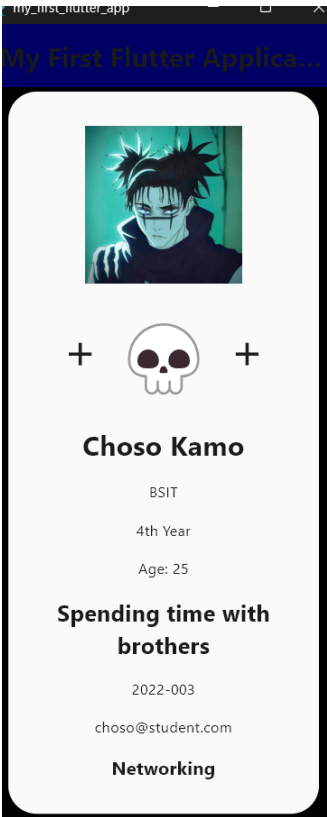
Sophia

Your UI should immediately reflect the new order.

Screenshot:

```
final sortedProfiles = [...profiles];

sortedProfiles.sort(
  (a, b) => (a.name ?? "").compareTo(b.name ?? ""),
);
```



🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Student List CTF.

Use an appropriate commit message following proper Git standards.

Branch:

feature/student-list

Screenshot:

Successful commit

Current branch

Terminal showing the commit

```
MINGW64:/c/IT314AB_Caballero

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-list)
$ git branch
  feature/data-ui
  feature/exam
  feature/exam_2
* feature/student-list
  feature/widgets
  main

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-list)
$ git status
On branch feature/student-list
Your branch is up to date with 'origin/feature/student-list'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   FlutterActivities/my_first_flutter_app/lib/main.dart
        modified:   FlutterActivities/my_first_flutter_app/linux/flutter/generated_plugin_registrant.cc
        modified:   FlutterActivities/my_first_flutter_app/linux/flutter/generated_plugin_registrant.h
        modified:   FlutterActivities/my_first_flutter_app/linux/flutter/generated_plugins.cmake
        modified:   FlutterActivities/my_first_flutter_app/macos/Flutter/GeneratedPluginRegistrant.swift
        modified:   FlutterActivities/my_first_flutter_app/windows/flutter/generated_plugin_registrant.cc
        modified:   FlutterActivities/my_first_flutter_app/windows/flutter/generated_plugin_registrant.h
        modified:   FlutterActivities/my_first_flutter_app/windows/flutter/generated_plugins.cmake

no changes added to commit (use "git add" and/or "git commit -a")

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-list)
$ git add
warning: in the working copy of 'FlutterActivities/my_first_flutter_app/linux/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterActivities/my_first_flutter_app/linux/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterActivities/my_first_flutter_app/linux/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterActivities/my_first_flutter_app/macos/Flutter/GeneratedPluginRegistrant.swift', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterActivities/my_first_flutter_app/windows/flutter/generated_plugin_registrant.cc', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterActivities/my_first_flutter_app/windows/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'FlutterActivities/my_first_flutter_app/windows/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-list)
$ git commit -m "feat: submit CTF 4"
[feature/student-list 9d3770d] feat: submit CTF 4
 1 file changed, 169 insertions(+), 216 deletions(-)

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-list)
$ git push
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 1.51 KiB | 774.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/mykyliecaballero/IT314AB_Caballero.git
   6bb4413..9d3770d  feature/student-list -> feature/student-list
```

